

TRACK 1 — AI & INTELLIGENT SOLUTIONS

Leverage ML and AI to improve social outcomes, optimise resource distribution, and democratise information access.

Challenge 1.1 — AI-Based Multilingual Chatbot for Welfare Scheme Awareness

Domain / Track: AI & Intelligent Systems

Background & Context India runs over 950 central and state welfare schemes — PM-KISAN, Ayushman Bharat, PMAY, Ujjwala, NREGA, Sukanya Samridhi, and hundreds of state-level entitlements. Yet a 2023 NITI Aayog field study estimated that fewer than 40% of eligible rural beneficiaries are aware of the schemes they qualify for, and an even smaller share complete the application successfully. The bottlenecks are not benefits design — they are information asymmetry, language barriers, low literacy, and intimidating documentation. Most government portals are English/Hindi-only, assume smartphone literacy, and break under low bandwidth.

Problem Statement Design a low-bandwidth, multilingual conversational assistant — deployable over WhatsApp, SMS, or IVR — that helps a rural user discover schemes they're eligible for, understand the documents required, and walk through the application process in their own language and dialect.

Why This Matters A single successful PM-KISAN enrolment is ₹6,000/year for a farming household; a successful Ayushman Bharat enrolment can prevent a medical-debt spiral. Closing the awareness gap is among the highest-leverage social interventions in India today — no new policy needed, only better last-mile delivery.

Project Objectives

- Build a working chatbot supporting at least two regional languages (e.g., Hindi + Tamil, or Bengali + Marathi).
- Cover a focused catalogue of 8–10 high-impact schemes rather than all 950.
- Design an eligibility-questioning flow that asks 4–6 simple questions and returns a personalised shortlist.
- Generate a downloadable/SMS-able checklist of required documents in the user's language.

Key Questions to Explore

- What is the lowest-friction channel for a user with a ₹500 keypad phone — WhatsApp, SMS, or IVR?

- How do you handle code-mixing (e.g., "मेरा aadhaar खो गया")?
- How do you keep the model honest when it doesn't know the answer, rather than hallucinating eligibility?
- What is the offline fallback when the user has no data balance?

Suggested Approach

- Use an LLM (Gemini, Llama 3, or an open Indic model like Sarvam-1 / IndicTrans2) behind a retrieval layer grounded in the official scheme database scraped from MyScheme.gov.in.
- Use RAG (retrieval-augmented generation) to prevent hallucinated entitlements — answers must cite the scheme.
- Build the front-end on Twilio WhatsApp Sandbox or Meta WhatsApp Business API (free tier).
- Compress the conversation flow into 4–6 turn dialogs to respect low bandwidth.

Recommended Tools & Resources

- APIs/Frameworks: Twilio, WhatsApp Business Cloud API, Rasa, LangChain, Hugging Face Transformers.
- Indic NLP: AI4Bharat IndicTrans2, IndicBERT, Bhashini.
- Datasets: MyScheme.gov.in scrape, data.gov.in scheme catalogues.
- Deployment: Replit, Render, or Hugging Face Spaces (free tiers).

Expected Deliverables

- Functional chatbot prototype (live demo link or video).
- User-flow diagram covering at least 3 personas (farmer, gig worker, woman head-of-household).
- Pilot test report with 10–15 real users from any nearby village or community.
- 2-page impact projection.

Success Metrics

- $\geq 80\%$ of test users complete the eligibility flow without dropping off.
- $\geq 70\%$ comprehension score on scheme summaries (measured via post-chat quiz).
- < 3 seconds median response time on 2G/3G simulation.

Potential Impact If adopted by a single district administration or NGO with 10,000 beneficiaries, even a 20% lift in scheme uptake translates to crores of rupees in entitlements reaching households that currently miss out.

Challenge 1.2 — Predictive Analytics for Resource Allocation in Disaster & NGO Operations